AI-Assisted CRISPR-Cas12a Guide RNA Design for Mycobacterium tuberculosis

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1. Introduction

Tuberculosis remains one of the world's leading infectious diseases. CRISPR-Cas12a has emerged as a promising tool for nucleic acid detection due to its high specificity and collateral cleavage activity. This project aims to develop a computational workflow for identifying and optimizing Cas12a guide RNAs targeting the MTB genome using bioinformatics and machine learning techniques.

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2. Objectives

- Identify Cas12a target sites
- Generate candidate guide RNAs
- Reduce off-target risk
- Apply machine learning for guide prioritization
- Optimize guides using biological design criteria

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3. Methodology

The MTB H37Rv genome was analyzed to identify TTTN PAM sequences. Twenty-base-pair guide RNAs were extracted and evaluated based on GC content. A simplified off-target analysis compared guide similarity against reference human and *E. coli* sequences. A Random Forest classifier ranked candidate guides using biological features. Finally, literature-inspired optimization criteria—including GC content, homopolymer detection, Poly-T detection, and Shannon entropy—were applied to generate the final guide recommendations.

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4. Results

The pipeline successfully identified candidate Cas12a guides and ranked them through a multi-stage optimization process. Machine learning provided predicted guide quality scores, while biological optimization further refined the ranking by eliminating guides with unfavorable sequence characteristics. The final output consists of prioritized guide RNA candidates suitable for future experimental validation.

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5. Discussion

Combining bioinformatics with machine learning improved guide prioritization beyond rule-based scoring alone. The project demonstrates how computational methods can support CRISPR assay design while highlighting the importance of biological constraints in guide selection.

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6. Limitations

- Simplified off-target analysis
- No RNA secondary structure prediction
- No experimental validation
- Machine learning trained on project-derived labels rather than experimental efficiency data

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7. Future Work

Future improvements include integrating genome-wide off-target search tools (e.g., Bowtie or Cas-OFFinder), RNA secondary structure prediction, experimentally validated Cas12a efficiency models, and laboratory validation of predicted guide RNAs.

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8. References

Singh, I., Compiro, P., Keawsapsak, P., & Hannanta-anan, P. (2026). *Deep Learning–Assisted Digital Microfluidic Platform for Automated CRISPR/Cas12 Detection of Mycobacterium tuberculosis*

Compiro et al. (2025): *CRISPR-Cas12a-Based Detection and Differentiation of Mycobacterium Spp*

Youngquist et al. (2025): *Rapid Tuberculosis Diagnosis from Respiratory or Blood Samples by a Low Cost, Portable Lab-in-Tube Assay*

Soh et al. (2022): *CRISPR-Based Systems for Sensitive and Rapid on-Site COVID-19 Diagnostics*

- Cas12a guide design
- DeepCas12 / CRISPR-DT (if cited)
- MTB reference genome (H37Rv)
- Biopython
- Scikit-learn